

Security and Privacy in Machine Learning

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Sharif University of Technology
Computer Engineering Department (CE)
Trustworthy and Secure AI Lab (TSAIL)



October 27, 2026

Today's Agenda

1 Course logistics

2 Security and Privacy

3 Deep Neural Networks

4 Security and Privacy in Machine Learning

Course logistics

Course information

- Course Number: 40816-1
  - Time: Sun. and Tues. 10:30 - 12:00
  - Rooms: CE-102 and https://vc.sharif.edu/ch/amsadeghzadeh
- Instructor
  - Amir Mahdi Sadeghzadeh (amsadeghzadeh@gmail.com)
    - Office: CE-718
    - Lab: Trustworthy and Secure AI Lab (TSAIL) CE-503
    - Office hours: by appointment and through email
- Course website: https://trustworthy-and-secure-ai-lab.github.io/SPML2026/
  - Syllabus, Lecture slides, Assignments, etc
- Quera: Quera page
  - Discussions and HWs
- TAs
  - Arian Komaei Koma (Head Course Assistant) - Email: ariankomaei@gmail.com
  - javad Hezareh (Head Course Assistant) - Email: hezarej@gmail.com

References

- No required textbook. This field moves faster than textbooks do. Every reading is a research paper. The main references for the course are many research papers in top-tier conferences and journals in computer security (SP, CCS, USENIX Security, EuroSP) and machine learning (NeurIPS, ICLR, ICML, CVPR, ECCV).

- The following books are used for reviewing background topics in machine learning and deep learning in the first part of the course.
  1. Christopher M. Bishop, Pattern Recognition and Machine Learning, Springer, 2006.
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  3. Ian Goodfellow, Deep Learning, MIT Press, 2016.

- Information security conferences
  1. IEEE Symposium on Security and Privacy (S&P)
  2. ACM Symposium on Computer and Communications Security (CCS)
  3. USENIX Security Symposium
  4. Network and Distributed System Security Symposium (NDSS)

- Machine learning conferences
  1. International Conference on Learning Representations (ICLR)
  2. Neural Information Processing Systems (NeurIPS)
  3. International Conference on Machine Learning (ICML)
  4. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)

Course objective

- Introduce students to the principals of the security and privacy in machine learning.
- The students become familiar with the vulnerabilities of machine learning in training and prediction phases and the methods to improve the robustness and privacy of machine learning.
- The course prepares students to explore new research directions and applications in the security and privacy of machine learning.

Pre-requisite

- **Deep Neural Networks**
- Basic Calculus and Linear Algebra
- Probability and Statistics
- Python (Numpy)
- Pytorch, Tensorflow

THIS IS NOT AN INTRODUCTORY COURSE!

- DO **NOT** TAKE THIS COURSE IF YOU DO NOT ALREADY HAVE HANDS-ON KNOWLEDGE OF MODERN ML TECHNOLOGIES.



Grading Policy

- Assignments (25%)
- Project (25%)
- Mid-term (20%)
- Final (30%).

Homeworks

- There are 4 or 5 homeworks

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- Alternative ways of completing homeworks
  - On local machines
  - On Google Colab
  - Kaggle
  - etc.

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- Ethics statement
  - Please read Sharif CE Department Ethics Statement
  - Every student must solve every homework by themselves
    - You may discuss the homeworks with your friends, but when you finally solve it, every line of your code (except libraries that have been okayed by course staff) must be written by you
    - **Your solution must be yours**

Homeworks

- **Important:**
  - All theory homework must be handwritten.
  - Do not submit LaTeX or Word-formatted documents for theory assignments — these will not be accepted.
  - You may use AI tools for assistance (e.g., brainstorming ideas, clarifying concepts, or debugging), but you must not include AI-generated text directly in your submission. All submitted work should be written in your own words and understanding.

Course Project Overview

Phase 1 — Exploring the Literature

- You'll work in teams of **3–4 students**, choosing a topic from those suggested by the teaching assistants (TAs).
- Each team will explore **at least 10 papers**, with **4–5 selected by the TAs** to help you get started. We'll also try to arrange a mentor for each topic.
- You'll share your review in a team presentation on a date to be announced.

Phase 2 — Bringing Ideas to Life

- Each team will implement **at least one paper** from Phase 1 and submit a report explaining how to run it and what results you obtained.
- You'll then try out your own ideas for improving the approach. **Thoughtful, well-documented attempts earn full credit, even if they don't lead to an improvement.**
- If your ideas do lead to an improvement, you'll earn **bonus credit**, with the amount determined by the instructor.

Ethical Statement

- This course covers topics in personal and public privacy and security. As part of this investigation we will explore technologies whose abuse may infringe on the rights of others.
- As instructor, We rely on the ethical use of these technologies. Unethical use may include circumvention of existing security or privacy measurements for any purpose, or the dissemination, promotion, or exploitation of vulnerabilities of these services. Exceptions to these guidelines may occur in the process of reporting vulnerabilities through public and authoritative channels.
- Any activity outside the letter or spirit of these guidelines will be reported to the proper authorities and may result in dismissal from the class.
- When in doubt, please contact the course professor for advice. **Do not** undertake any action which could be perceived as technology misuse anywhere and/or under any circumstances unless you have received explicit permission from Dr. Sadeghzadeh.

(Papernot, 2019)

Course outline

1. Security of Machine Learning Models

- Adversarial examples: iterative, physical, universal, and black-box attacks
- Empirical defenses, certified robustness, and data poisoning

2. Privacy and Model Protection

- Model extraction, intellectual property, and watermarking
- Membership and attribute inference attacks
- Differential privacy, private deep learning, and machine unlearning

3. Safety and Security of LLMs and AI Agents

- Alignment, RLHF, jailbreaking, and red-teaming
- Agent threat modeling, prompt injection, and memory poisoning
- Tool authorization, multi-agent security, and defenses

!



Welcome to SPML - Autumn 2026 :)

Security and Privacy

Security

- Achieving some goal in the presence of an adversary.

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  - **Policy**: the goal you want to achieve.
    - e.g. only Alice should read file F.
    - Common goals: Confidentiality, Integrity, Availability (CIA), and privacy.

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  - Result: no way for adversary within threat model to violate policy

(Zeldovich, 2014)

Confidentiality and Privacy

- **Confidentiality**: the property that information is not made available or disclosed to unauthorized individuals, entities, or process (i.e. to any unauthorized system entity). [definitions from RFC 2828]
  - Not the same as privacy

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- **Privacy**: the right of an entity (normally a person), acting in its own behalf, to determine the degree to which it will interact with its environment, including the degree to which the entity is willing to share information about itself with others.
  - Privacy is a reason for confidentiality

(Kharrazi, 2015)

Integrity

- **Data integrity**: The property that data has not been changed, destroyed, or lost in an unauthorized or accidental manner.
  - Often of more commercial interest than confidentiality
- **System integrity**: The quality that a system has when it can perform its intended function in an unimpaired manner, free from deliberate or inadvertent unauthorized manipulation.

(Kharrazi, 2015)

Availability

- **Availability**: The property of a system or a system resource being accessible and usable upon demand by an authorized system entity, according to performance specifications for the system i.e. a system is available if it provides services according to the system design whenever users request them.
  - Denial of service attacks are direct assaults on availability
  - Turning off a computer provides confidentiality and integrity, but hurts availability

(Kharrazi, 2015)

More Definitions

- **Vulnerability**: An error or weakness in the design, implementation, or operation of a system.
- **Attack**: A means of exploiting some vulnerability in a system.
- **Threat**: An adversary that is motivated and capable of exploiting a vulnerability. The violation need not actually occur for there to be a threat.

(Kharrazi, 2015)

Why is security hard?

- It's a negative goal.
  - It is easy to check whether a positive goal is upheld.
  - E.g., Alice can actually read file F.
- Harder to check that there's no possible way Alice can read file F.
  - How would you even begin to enumerate all the possible ways Alice could go about reading the file?
  - Too many layers at which Alice could exploit bugs to gain access to file F.
- Difficult to think of all possible ways that attacker might break in.
- Realistic threat models are open-ended.
  - Iterative process: design, update policies, threat model, and Mechanisms as necessary.
  - The weakest link matters.

(Zeldovich, 2014)

Why is security hard?

- In this class, we'll push the boundary of each system to see when it breaks.
- Each system will likely have some breaking point.
- Doesn't necessarily mean the system is not useful. It depends on the context.
- Important to understand what a system can do, and what a system cannot.

(Zeldovich, 2014)

The Art of War

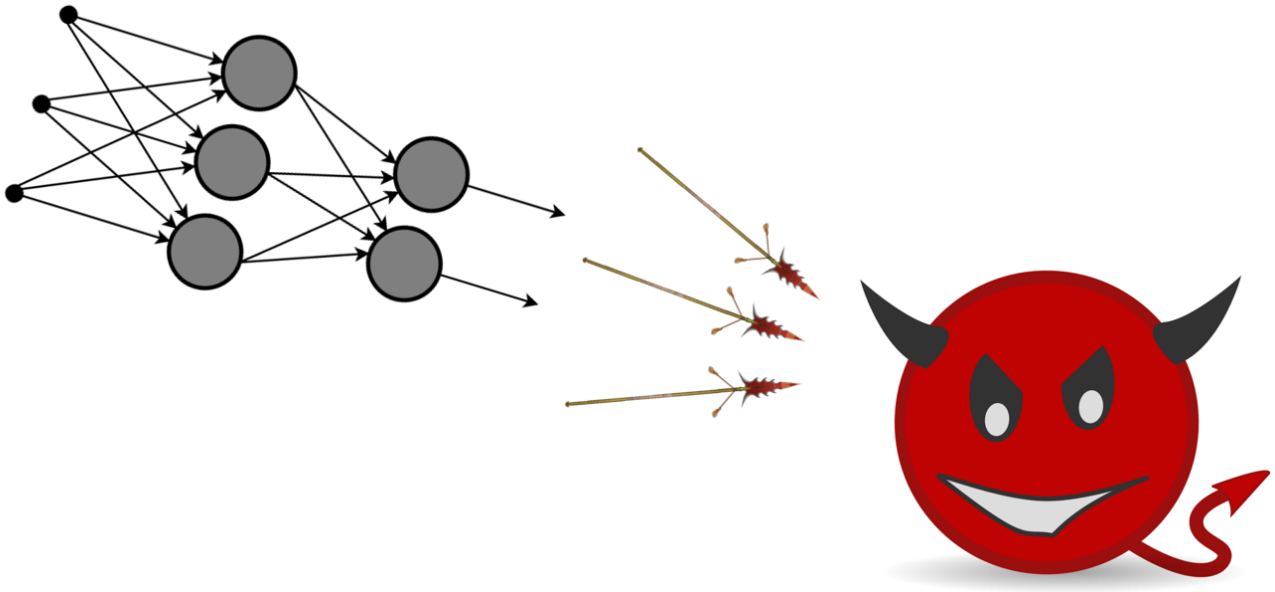
- If you know the enemy and know yourself, you need not fear the result of a hundred battles.
- If you know yourself but not the enemy, for every victory gained you will also suffer a defeat.
- If you know neither the enemy nor yourself, you will succumb in every battle.



Sun Tzu, roughly 5th century BC

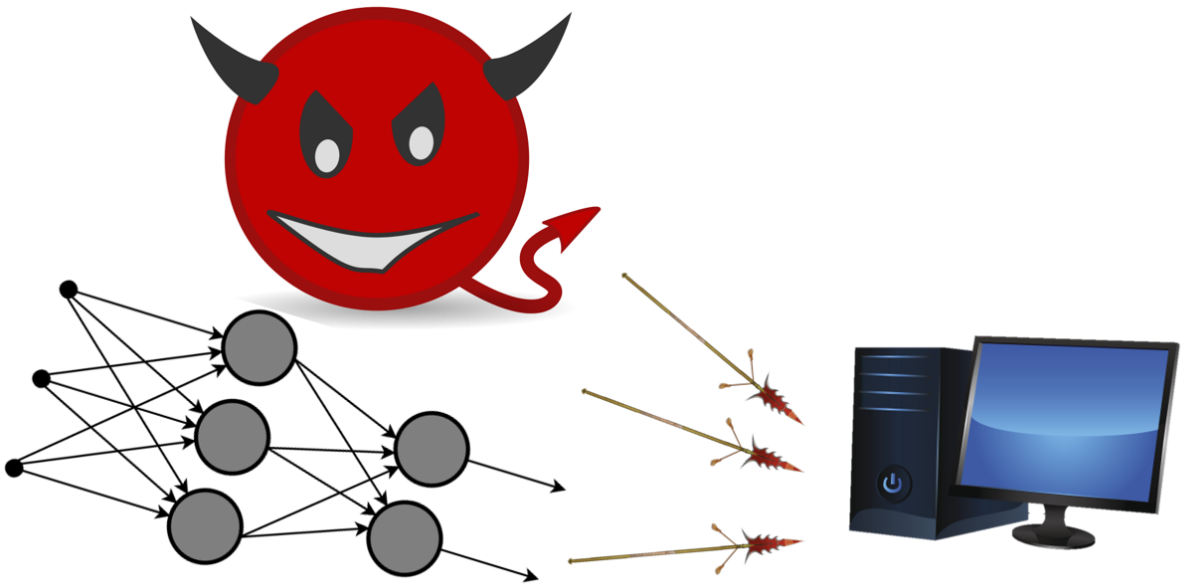
Machine Learning and Security

- Machine Learning for Security
  - Malware detection
  - Intrusion detection
  - Malicious network traffic detection



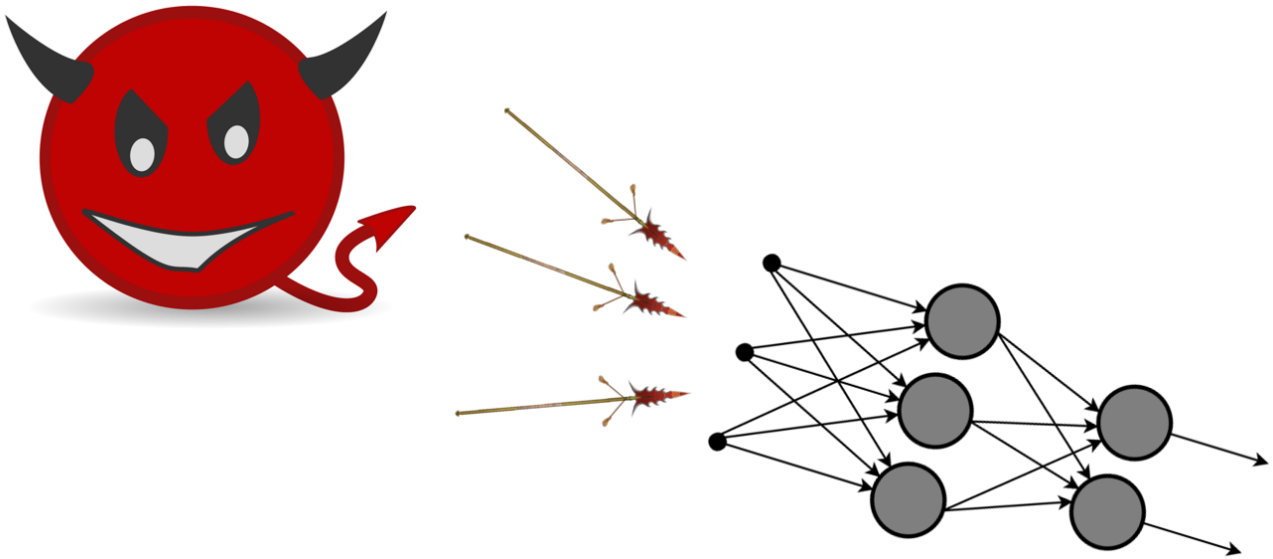
Machine Learning and Security

- Security against Machine Learning
  - Password guessing
  - Fake reviews
  - Website fingerprinting
  - Sandbox evading



Machine Learning and Security

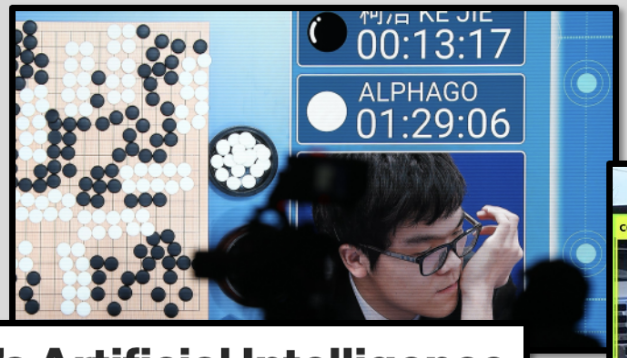
- Security of Machine Learning
  - Adversarial example
  - Poisoning model
  - Membership inference attacks
  - Model extraction

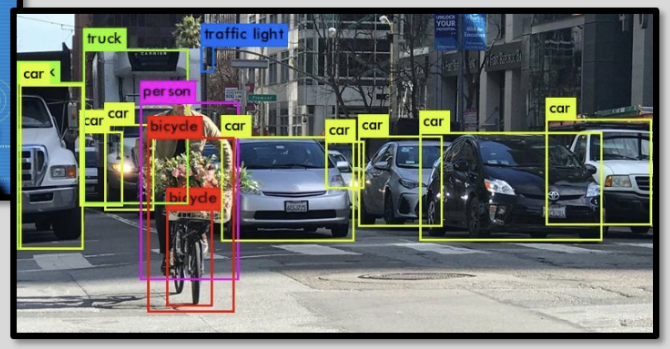


Deep Neural Networks

Deep Neural Networks are taking over!







Google's Artificial Intelligence Built an AI That Outperforms Any Made by Humans

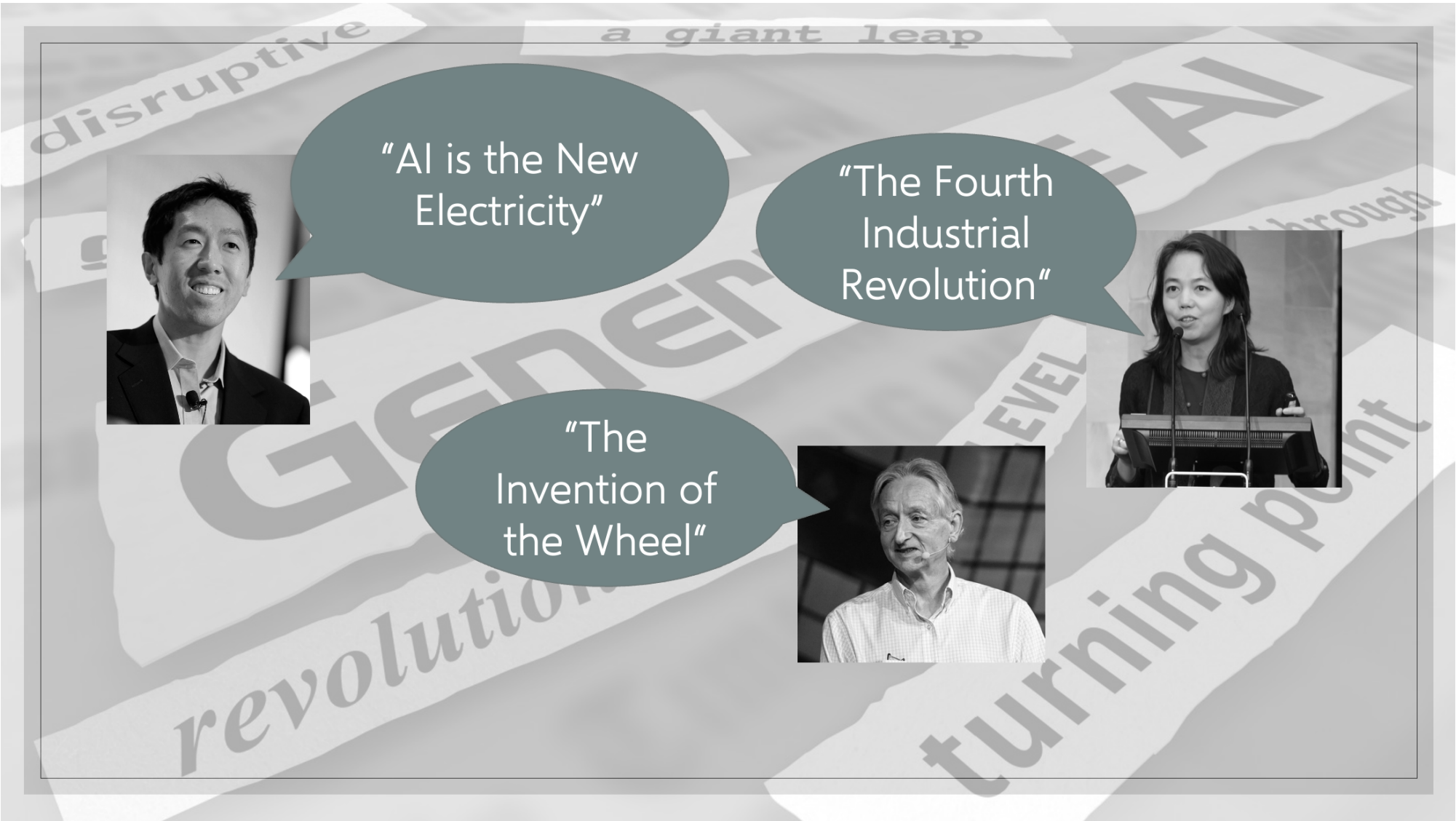
Research Highlight | Publish

BREAST CANCER

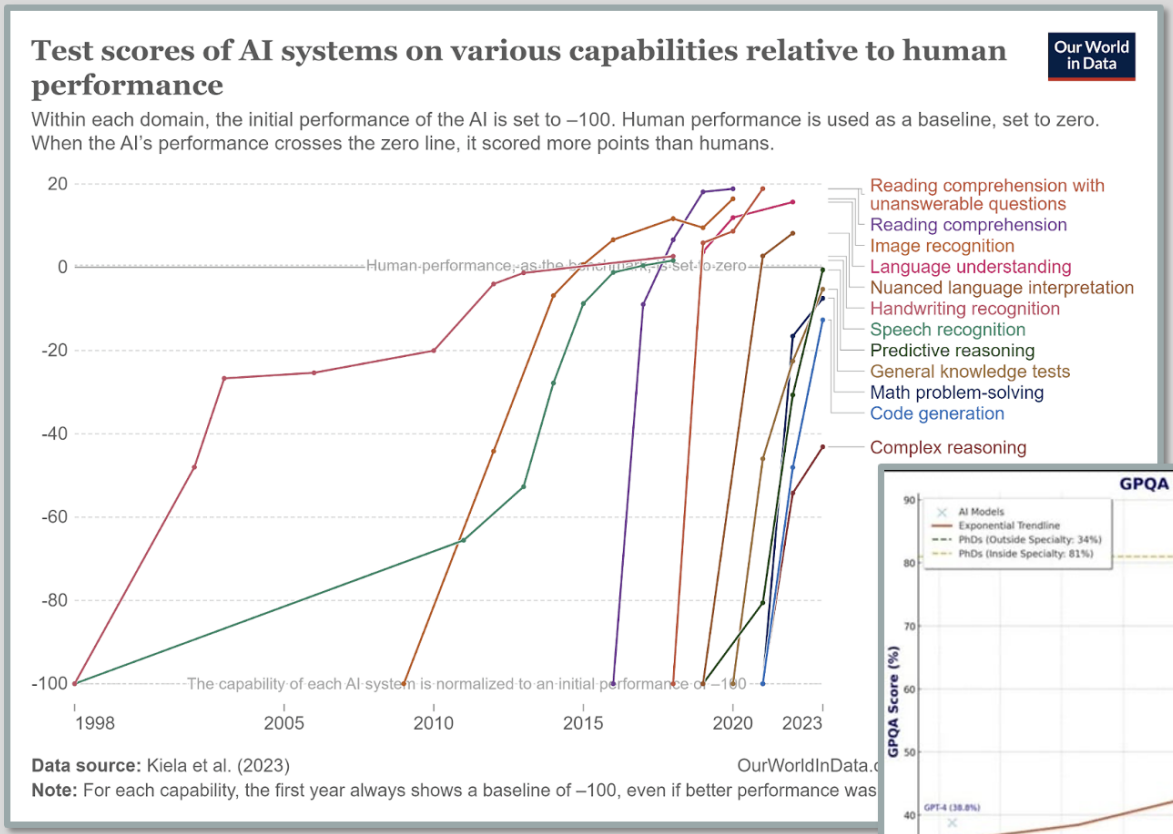
AI outperforms radiologists in mammographic screening

Microsoft's Deep Learning Project Outperforms Humans In Image Recognition

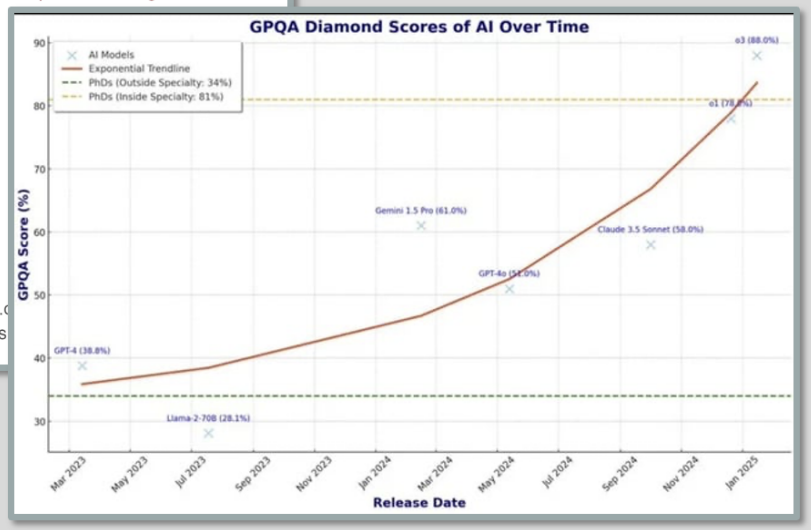
Deep Neural Networks are taking over!



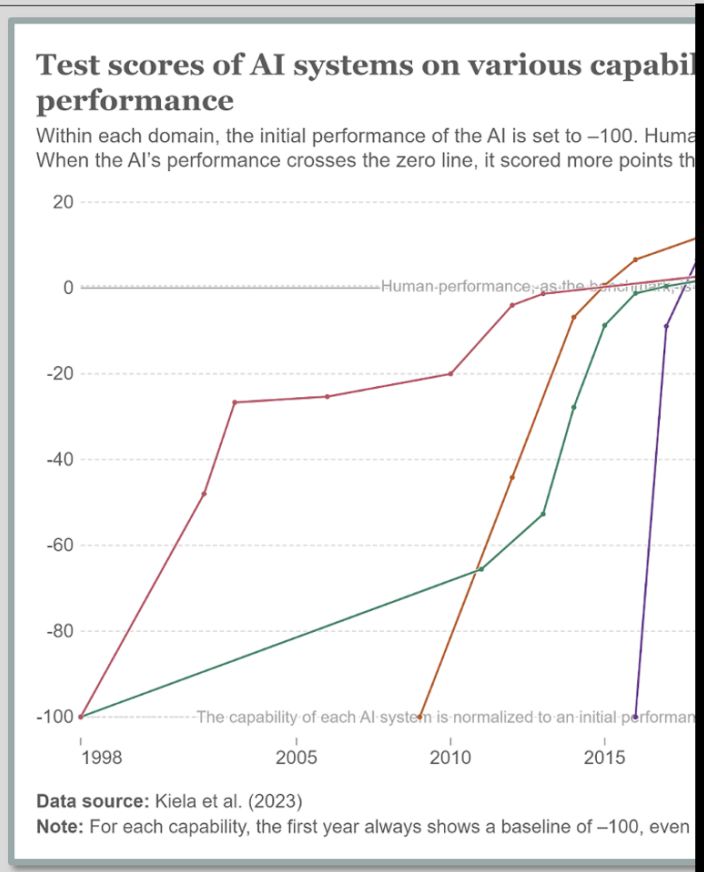
Deep Neural Networks are taking over!





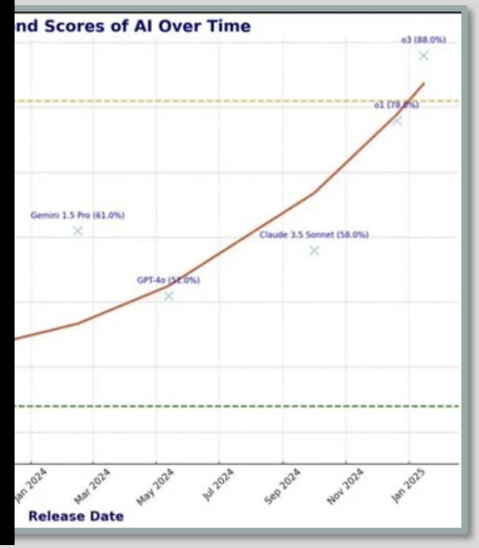


Deep Neural Networks are taking over!









Deep Neural Networks (DNNs)

- Learning a computational model consists of multiple processing layers
  - learn representations of data with multiple levels of abstraction.
- Dramatically improved the state-of-the-art in many vision, NLP, and speech tasks (and also in many other domains like bioinformatics, malware detection, intrusion detection, etc.)



(LeCun, 2020)

Supervised learning

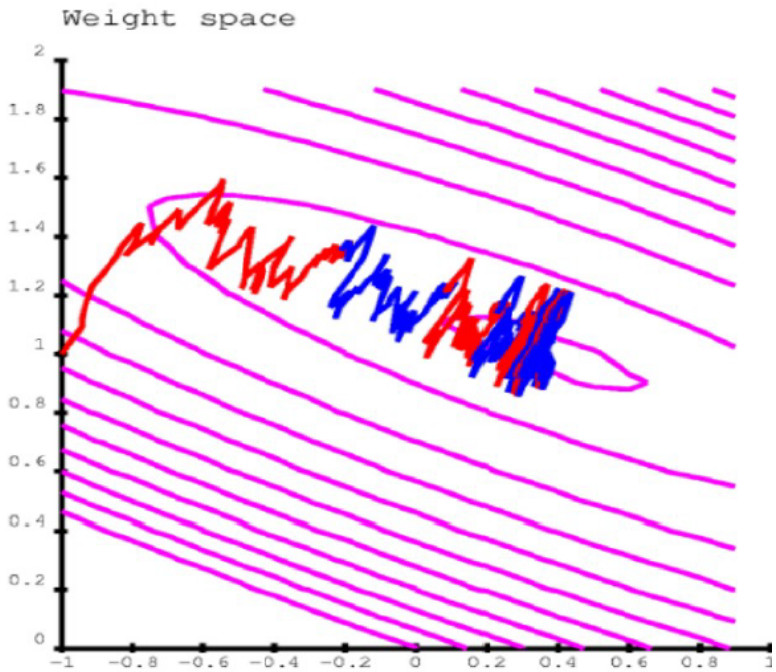


Function with adjustable parameters



Objective Function

traffic light: -1

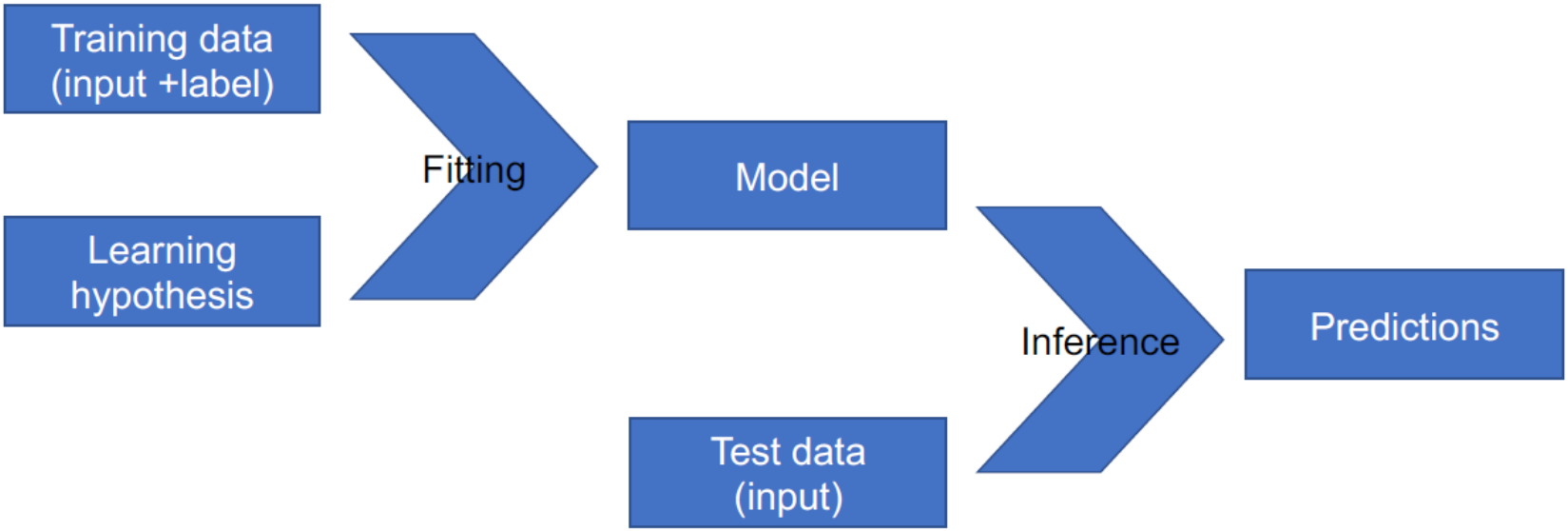


$$W_i \leftarrow W_i - \eta \frac{\partial L(W, X)}{\partial W_i}$$

(LeCun, 2020)

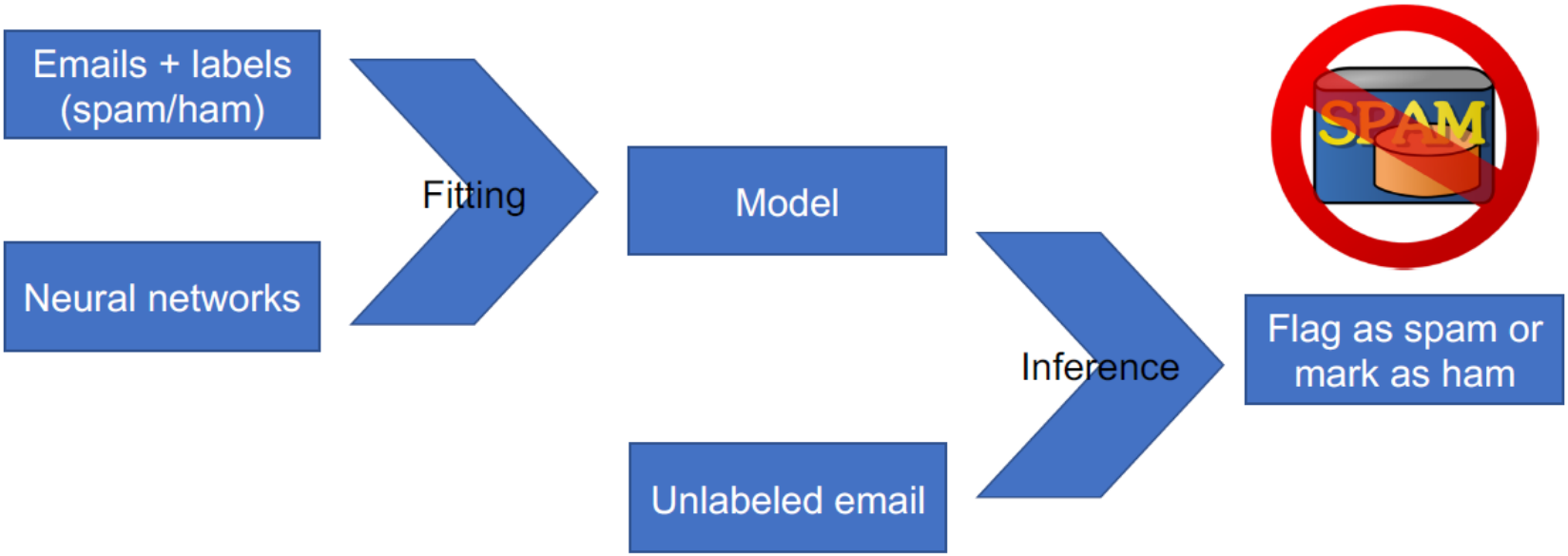
Security and Privacy in Machine Learning

Machine learning (supervised learning) paradigm



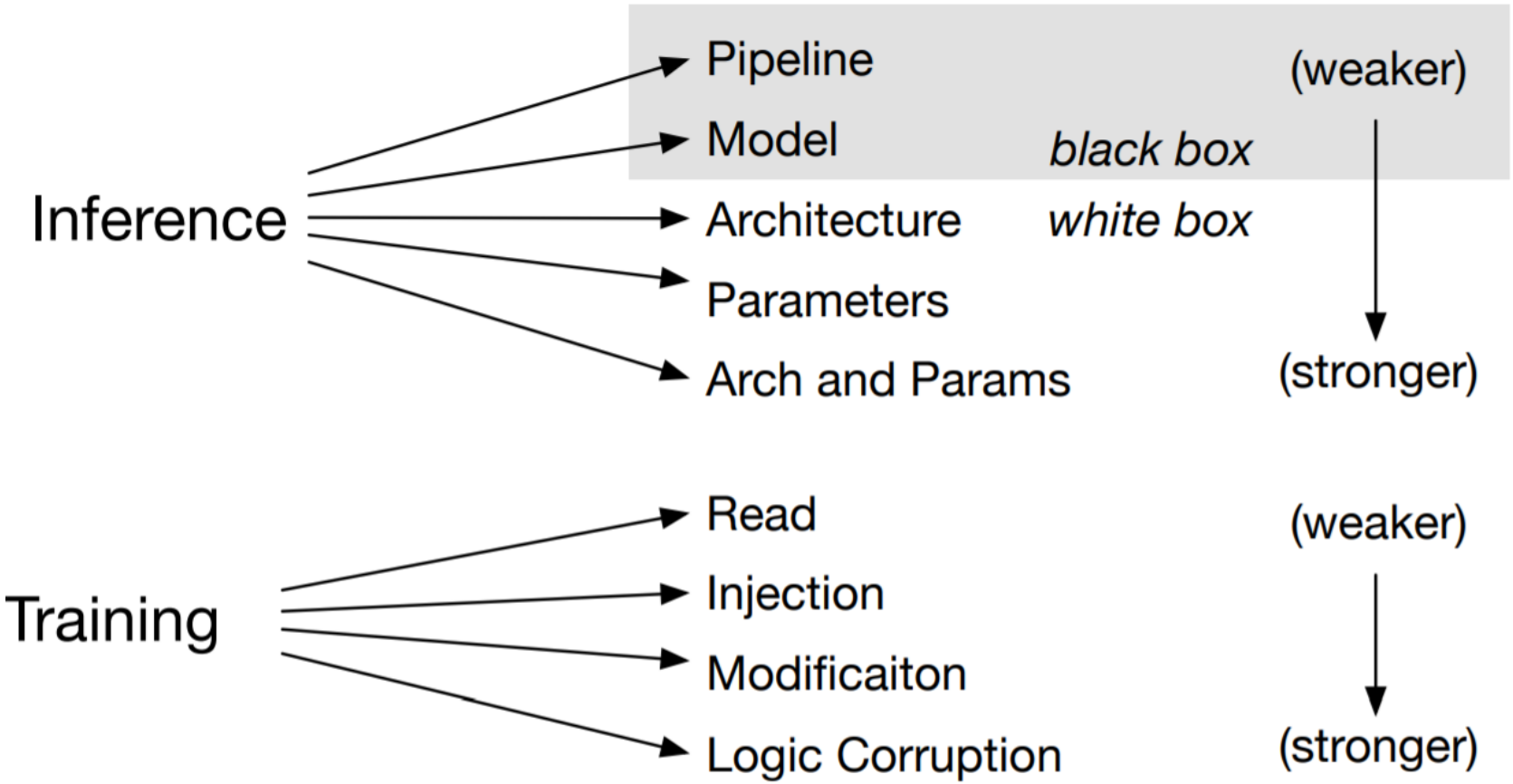
(Papernote, 2019)

ML for spam detection



(Papernote, 2019)

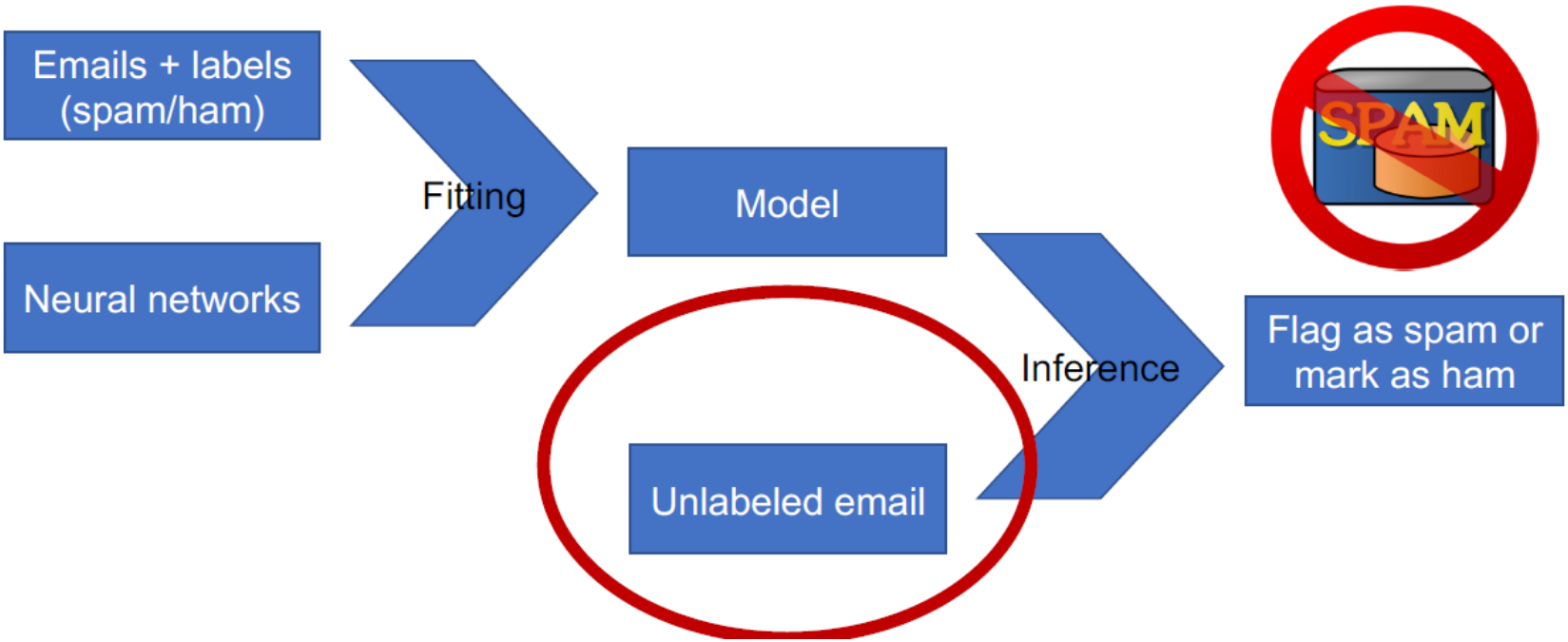
Threat Model (Adversarial Capabilities)



(Papernote et al, 2016)

Evasion

- Adversary crafts adversarial example that evades detection (spam email instantly marked as ham)
  - Goal: violation of integrity.
  - Threat model: adversary can send a request to the model.
  - Mechanism: adversarial training.



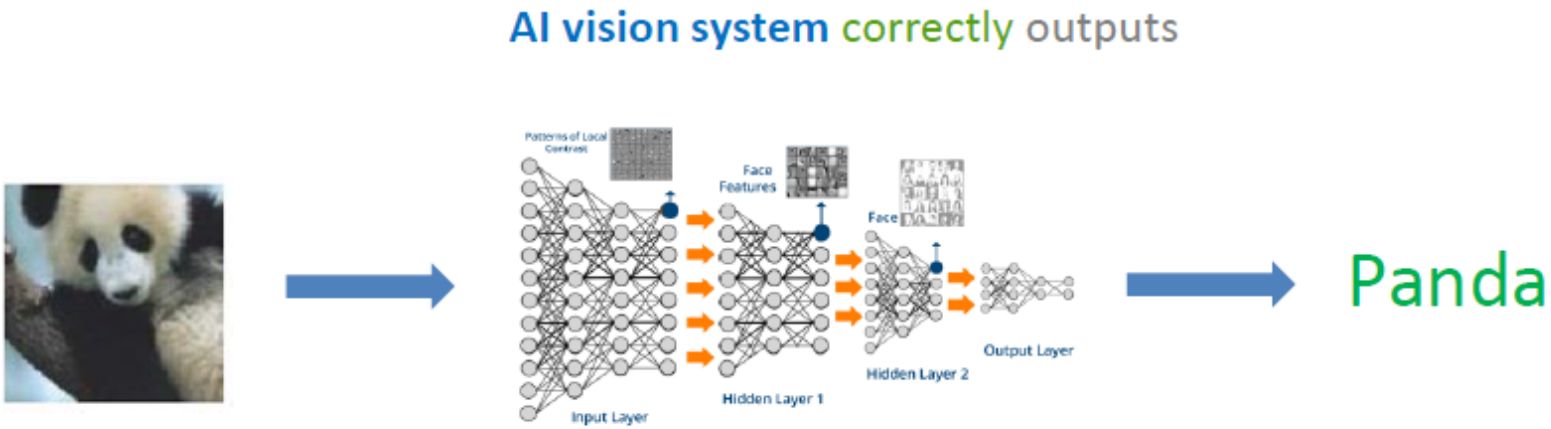
(Papernote, 2019)

Evasion

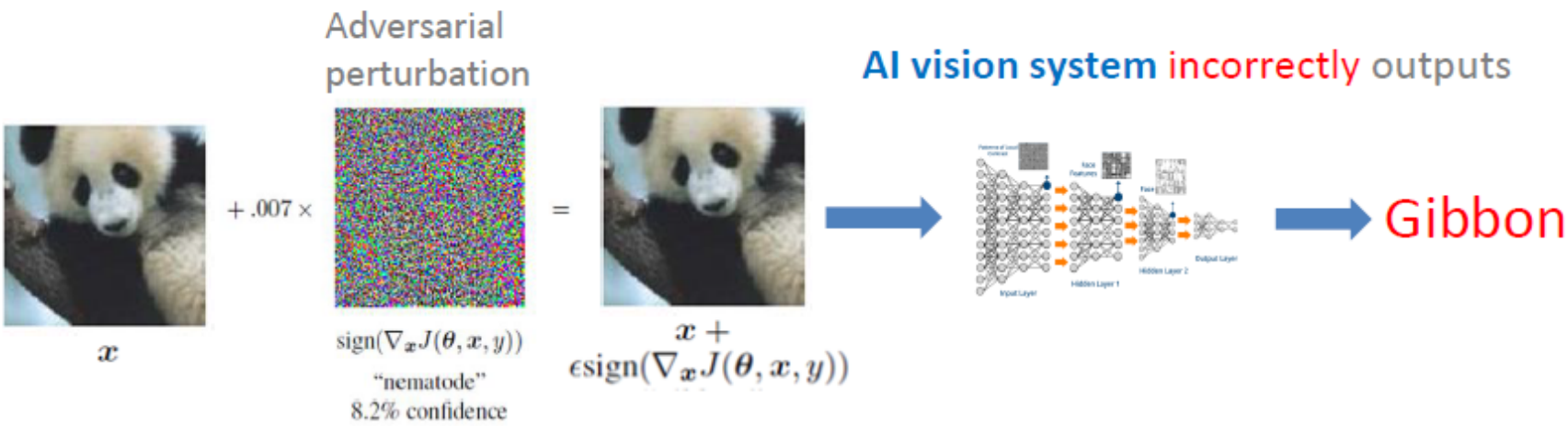


What animal do **you** see?

Evasion


AI vision system correctly outputs
Patterns of Local Contrast
Face Features
Face
Input Layer
Hidden Layer 1
Hidden Layer 2
Output Layer
Panda

Evasion



Explaining and Harnessing Adversarial Examples, ICLR '15

Evasion

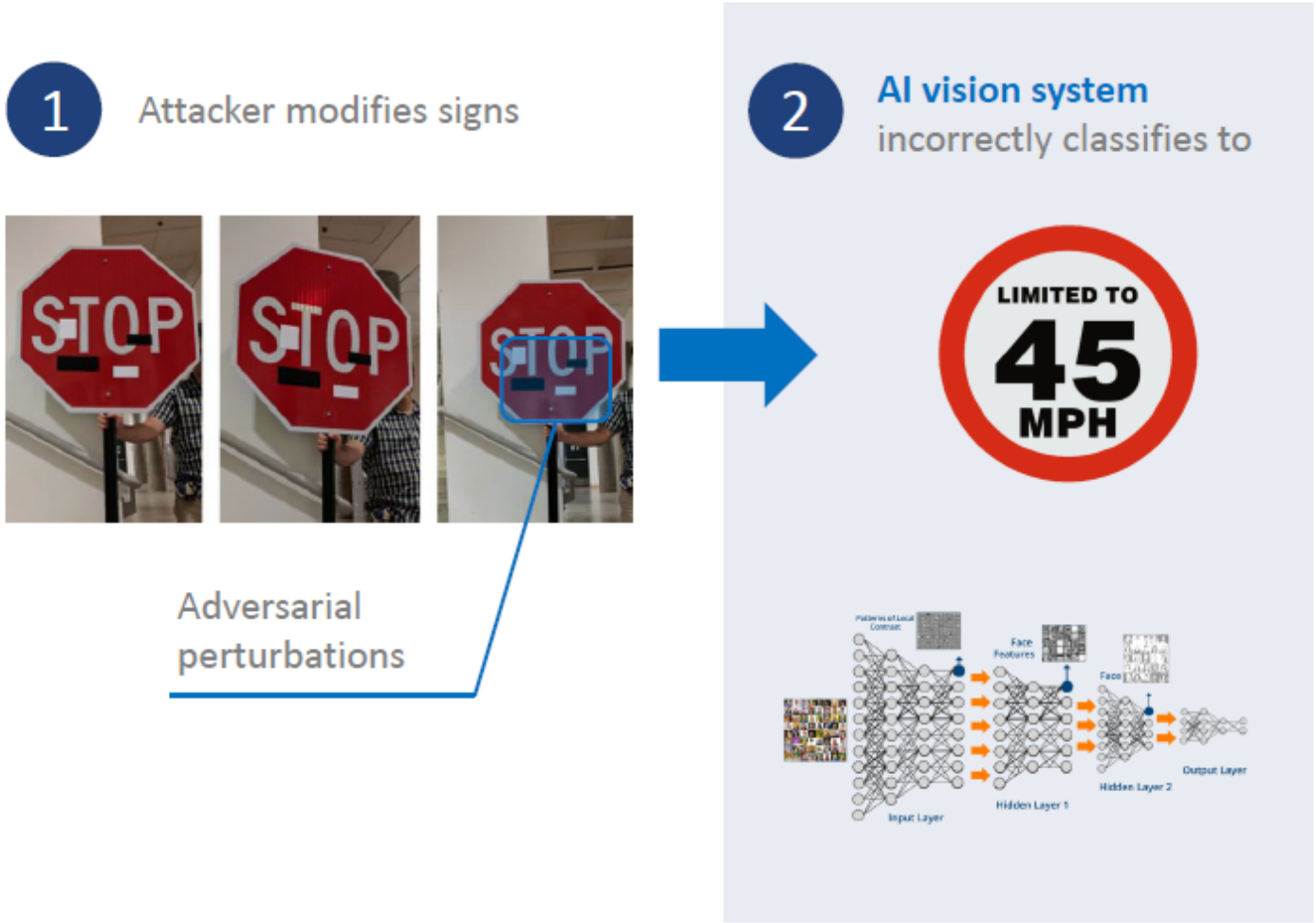
1 Attacker modifies signs



What sign do **you** see?

Robust Physical-World Attacks on Deep Learning Visual Classification, CVPR'18

Evasion



Robust Physical-World Attacks on Deep Learning Visual Classification, CVPR'18

Evasion

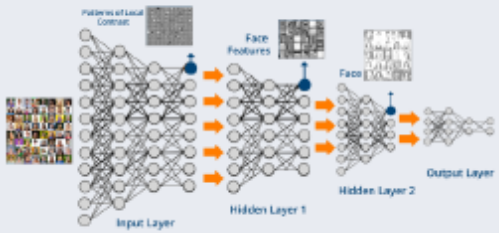
1 Attacker modifies signs



Adversarial perturbations

2 AI vision system incorrectly classifies to





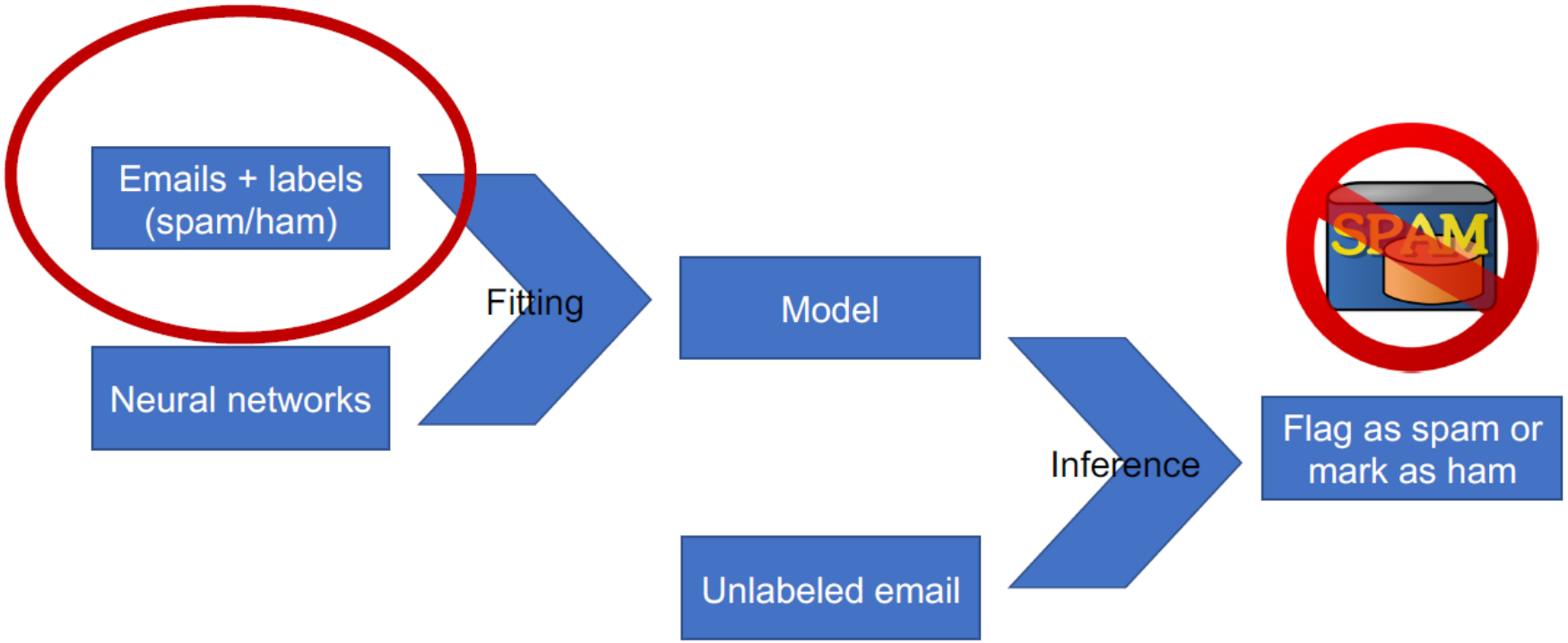
3 Car crash



Robust Physical-World Attacks on Deep Learning Visual Classification, CVPR'18

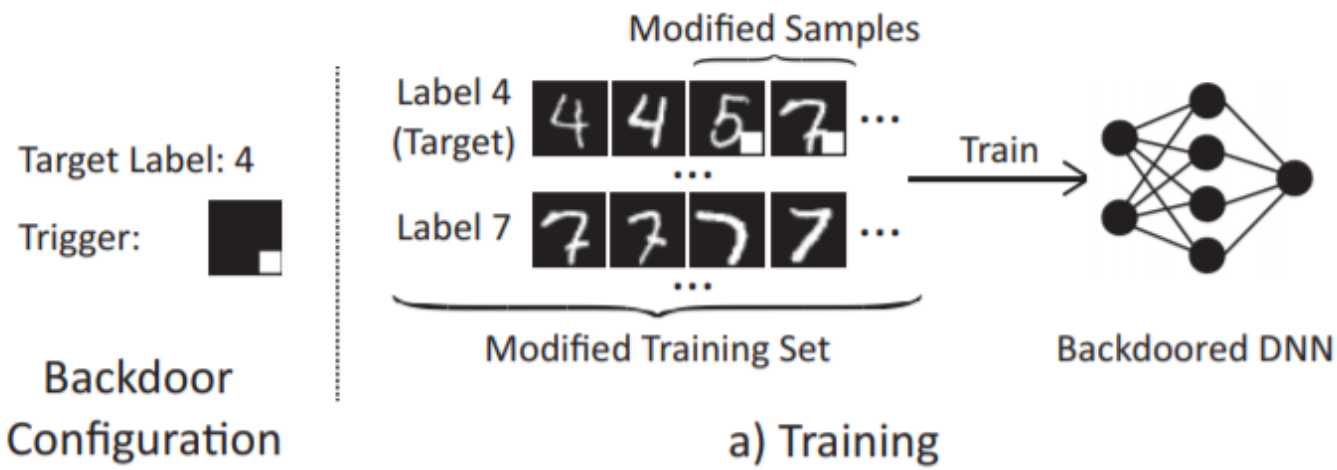
Poisoning

- Adversary inserts emails that contain spam but removes them from the spam folder back to inbox.
  - Goal: violation of integrity
  - Threat model: the adversary can modify the training set.
  - Mechanism: detecting anomaly training data

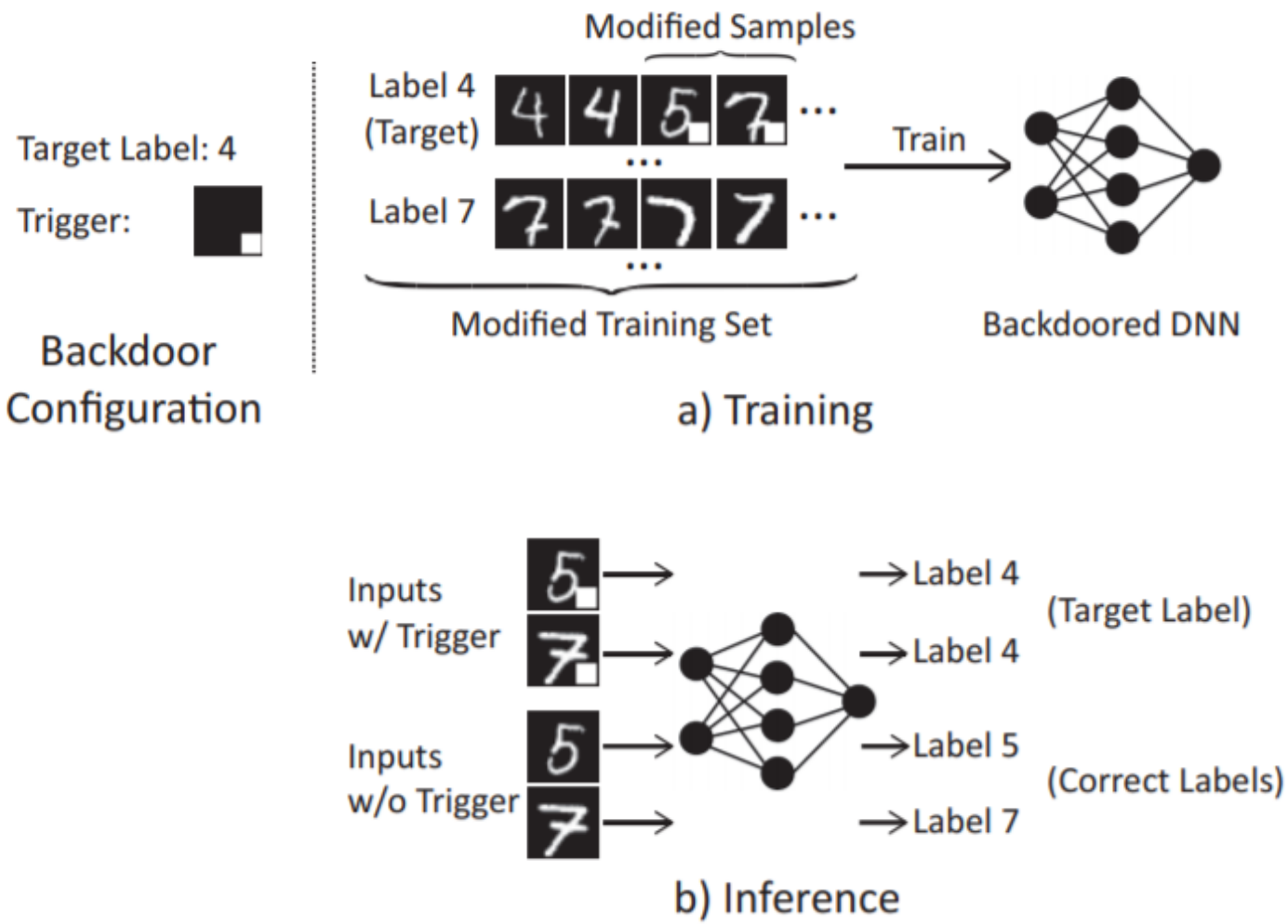


(Papernote, 2019)

Poisoning


Target Label: 4
Trigger:
Backdoor Configuration
Modified Samples
Label 4 (Target)
Label 7
Modified Training Set
Train
Backdoored DNN
a) Training

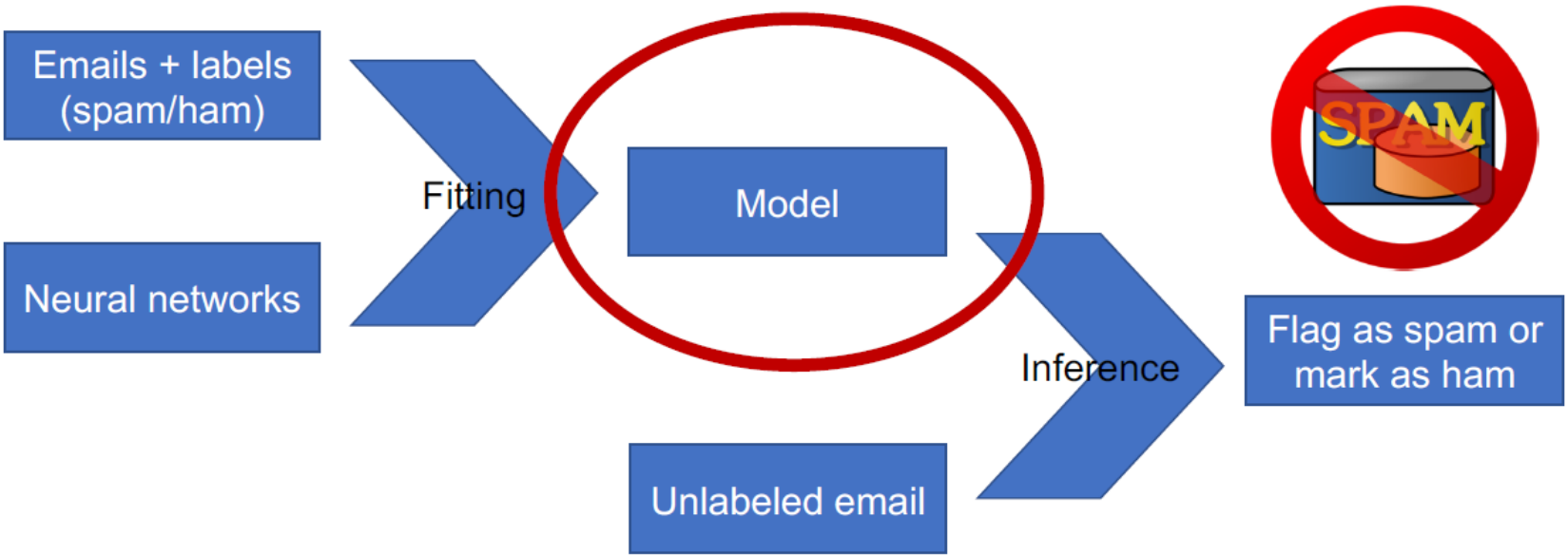
Poisoning



(Wang et. al., 2019)

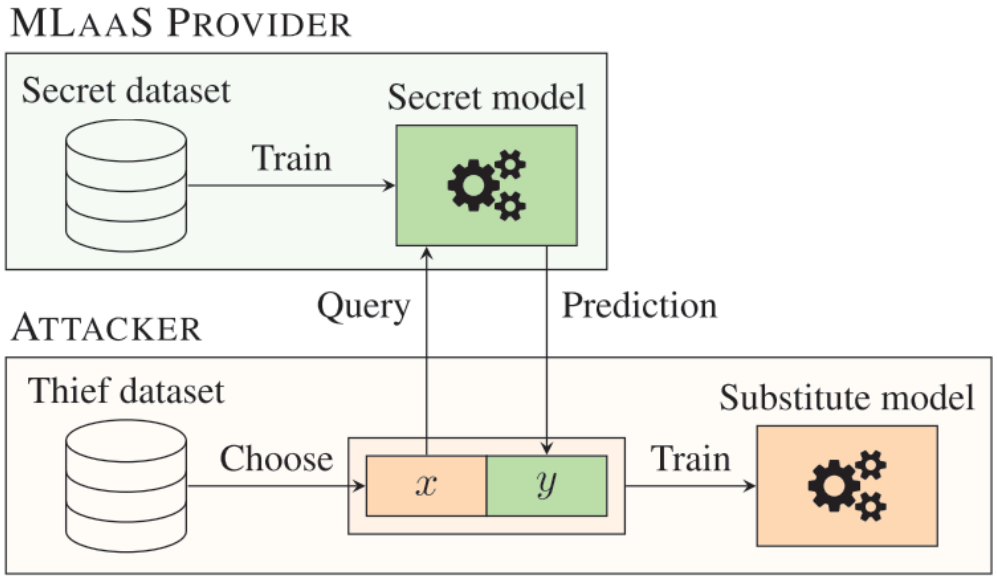
Model extraction

- Adversary observes predictions and reconstructs model locally
  - Goal: violation of privacy (confidentiality).
  - Threat model: adversary has access to the input and the output of model.
  - Mechanism: detecting sample sequences of attack



(Papernote, 2019)

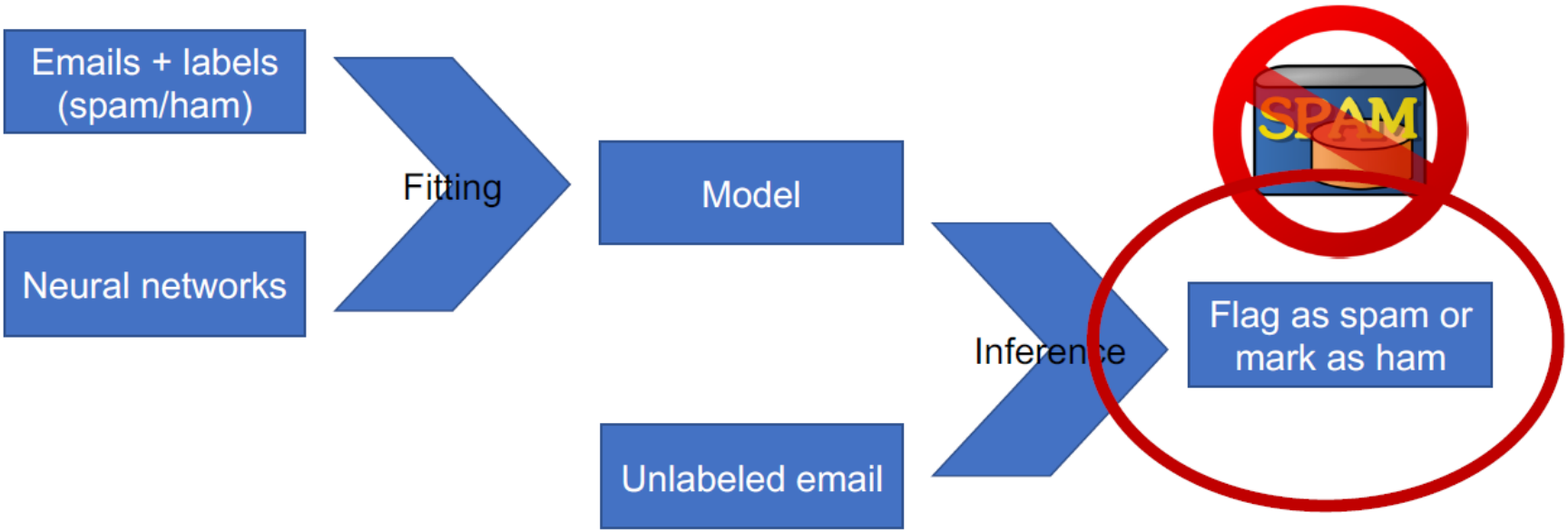
Model extraction



(Soham Pal, 2020)

Membership inference attack

- Adversary inspects model to test whether an email was used to train it (privacy violation)
  - Goal: violation of privacy (confidentiality).
  - Threat model: adversary can send a request to the model.
  - Mechanism: Differential Privacy.



(Papernote, 2019)

Membership inference attack

